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Tradesecrets - Instrumentation and Control Technician

Government of Alberta

8-10 minutes

Duties

Instrumentation and control technicians install, maintain and repair the measuring and control devices used in industrial and commercial processing.

Instrumentation and control technicians work with a wide variety of pneumatic, electronic and microcomputer devices used to measure and control pressure, flow, temperature, level, motion, force, and chemical composition. In general, they:

- consult manufacturers' manuals to determine test and maintenance procedures,
- use pneumatic, electrical and electronic testing devices to inspect and test instrument and system operation, and diagnose faults,
- practice loss management principles,
- consult with and advise process operators,
- train apprentices,
- repair, maintain and adjust system components or remove and

replace defective parts,

- do risk assessments,
- calibrate and maintain components and instruments according to manufacturers' specifications,
- work with engineers on basic design,
- install and maintain instruments on new or existing plant equipment and processes, and
- interpret and use appropriate CSA, ISA and API installation standards and practices.

For more information regarding tasks and activities, please review the Trade Regulation.

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Working Conditions

Instrumentation and control technicians may work regular week day hours or shifts. They can work various rotation schedules (such as 7 days on and 7 days off). They are sometimes on-call at night and on weekends. Working conditions can change dramatically from one job to another. Instrumentation and control technicians working with manufacturing processes may be exposed to confined spaces, high elevations, noisy, dusty, cold or unusually warm conditions. There may also be exposure to radiation devices and laser equipment.

There are some safety hazards, particularly when processing dangerous chemicals or working with substances under pressure or at high temperatures. The track record for safety in this profession

is excellent with a low incident rate.

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Skills and Abilities

The work is most rewarding for those who enjoy security, problem-solving, developing an area of expertise and working with little direction or supervision.

To be successful, instrumentation and control technicians must have:

- the ability to pay careful attention to details,
- good communication and reading skills,
- manual dexterity and patience,
- the ability to use proper lifting techniques to lift between 11 and 25 kilograms,
- good mathematical, scientific, mechanical and logical reasoning abilities, and
- the ability and desire to keep up-to-date with technological developments in the field.

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Employment and Advancement

With industry becoming increasingly automated, instrumentation and control technicians are needed virtually anywhere there are control and metering systems. They are employed in the following industries:

- pulp and paper processing,
- hydro power generation,
- mining, petrochemical and natural gas,
- industrial and commercial manufacturing,
- industrial construction, and
- industrial instrument servicing.

Journeyperson wage rates vary, but generally range from $30 to $55 an hour, plus benefits.

Experienced instrumentation and control technicians may advance to supervisory positions, be employed as engineering technicians or move into company sales offices.



Working in Alberta

To work as a instrumentation and control technician in Alberta, a person must:

- be a registered apprentice, an Alberta-certified journeyperson, hold a valid recognized credential , OR
- work for an employer who is satisfied that the worker has the skills and knowledge expected of a certified journeyperson, OR
- be self-employed.

Individuals possessing a valid recognized credential in Alberta are eligible to receive a Blue Seal business credential after completing the necessary requirements.



Apprenticeship Training

The term of apprenticeship for instrumentation and control technicians is 4 years (four 12-month periods), including a minimum of 1500 hours of on-the-job training and 8 weeks of technical training in the first and second years, and a minimum of 1425 hours of on-the-job training and 10 weeks of technical training in the third and fourth years.

- An applicant who previously completed courses of study or work experience related to the instrument technician trade or holds a related journeyperson certificate and has the employer's recommendation, may qualify for credit that could reduce the term of apprenticeship. Inquiries about credit for previously completed courses of study or work experience can be directed to an apprentice representative toll-free at 1-800-248-4823.
- A person who has previous training or work experience in the trade and wants to determine their level of skill and knowledge for entry or advanced standing in an apprenticeship program may complete the online Prior Learning Assessment Application. For more information, see the online Prior Learning Assessment Guide.
- A high school student can become an apprentice and gain credits toward apprenticeship training and a high school diploma at the same time under the Registered Apprenticeship Program (RAP).

To learn the skills required of a instrumentation and control technician in Alberta and be issued an Alberta Journeyman Certificate, a person must:

- satisfy the entrance requirements of an Alberta high school diploma that includes English 20-2, Math 30-3, Physics 30 and Chemistry

30, or equivalent, or pass the entrance exam (see Entrance Level Competencies, Exam Counselling Sheet, Entrance Exam Study Guide, and Entrance Exam Support Materials List)

- find a suitable employer who is willing to hire and train an apprentice

  Most employers prefer to hire high school or post-secondary program graduates who have strong backgrounds in the chemical and physical processes involved in instrumentation.

  For holders of an existing trade certificate, it can serve as your entrance requirement equivalency when registering in an additional trade.

- complete the online Apprenticeship Application and Contract
- pay the non-refundable application fee as part of the application process
- complete the required on-the-job training
  - during on-the-job training, apprentice instrumentation and control technicians earn at least 55 percent of the journeyperson wage rate in their place of employment in the first year, 65 percent in the second, 75 percent in the third, and 85 percent in the fourth year.
- complete all program requirements as identified in the course outline
- enroll in technical training
- review books and materials required for training
- successfully complete all required exams

  Apprentices may attempt the Interprovincial Exam in the final period of their apprenticeship training and, if successful, be granted a Red

Seal.

Apprentices involved in the Career and Technology Studies (CTS) stream in high school may be eligible for credit towards their apprenticeship first period training.

When apprentices attend technical training, they are required to pay the applicable tuition fee and purchase course supplies.

Student loans, grants, scholarships and other financial assistance may be available. For more information see Financial Assistance, visit an Apprenticeship and Industry Training office or call toll-free to 1-800-248-4823.



Interprovincial Standards Red Seal Program

An instrumentation and control technician who holds a valid trade certificate from Alberta or from another Canadian province or territory may apply to write the Interprovincial Exam and, if successful, be granted a Red Seal under the Interprovincial Standards Red Seal Program. The Red Seal is recognized throughout most of Canada.

To prepare for the exam see the Exam Counselling Sheets and review available resource materials.



Qualification Certificate Program

For a Qualification Certificate based on a recognized credential or work experience in order to prepare for the exam(s) please refer to the Exam Counselling Sheets and review available

resource materials.

Time spent on supervisory or foreman duties, counter work, heading the tool crib, or on training course is NOT counted as 'hands-on' work experience.

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Equivalency Program

A person who holds a valid recognized credential does not require an Alberta equivalency document to work in the trade in Alberta.

However, some employers may require Alberta documentation as proof that the holder is allowed to work in the trade or that the holder's credential is recognized.

Click for more information on the Equivalency Program.

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